

BIG-BANG NUCLEOSYNTHESIS FROM HQIV NETWORK WEIGHTS: LOCK-IN η , EPOCH LADDER, AND INTEGRATED WITNESSES IN THE LEAN LIBRARY

STEVEN ETTINGER JR

ABSTRACT. **Position in the publication chain.** This is the eighth paper in the HQIV preprint series and the fifth Tier-2 closure note. Tier 0–1 fix the discrete spine [7, 9, 11, 13, 10, 8, 12]; Tier 2 supplies the published mass lock-in synthesis, baryogenesis normalisation, and nucleon composite-trace binding [6, 5, 4]. The present note states what the BBN Lean stack *actually proves* about primordial light-element readouts when lock-in η , nucleon masses, and composite-trace binding weights are already fixed upstream.

Result. At lock-in the library defines an 8×8 network readout for Y_p , D/H , ${}^3\text{He}/H$, and ${}^7\text{Li}/H$ from valley counts, Boltzmann weights, and cluster binding Q values tied to [6]. A separate epoch module places the BBN window on the temperature shell ladder between QCD lock-in and the CMB today; an integrated witness certifies Y_p in the observed band and within 1% of the freeze-out partition value at the same η . Machine-checked vital bundles aggregate positivity, ladder ordering, and η consistency ([full BBN vital readout](#)). Reader-friendly anchors with clickable Lean links are in Appendix A.

Single absolute anchor (strictly isolated). Dimensionful nucleon masses enter only through the proton lock-in witness at reference $M = 4$; lock-in η_{paper} is imported from the baryogenesis witness layer, not refit here. Coc et al. semi-analytic fits [1] are labelled *comparison-only* in Lean and are never HQIV inputs.

Patch-ontology reader contract. Per the patch-theory reader contract below, BBN is a discrete-epoch readout on the shell-temperature map, not a continuum-limit completion programme. Full PRIMAT-style reaction networks, Li-problem resolution, and sphaleron/chemical-potential transport remain comparison-layer or future work.

Status. Proved: network vital readout, epoch vital readout, integrated Y_p witness pack, umbrella `bbn_full_vital_readout`, and the stoichiometric integrator spine ([BBN stoichiometric integrator](#)). Python reproduction at lock-in η_{paper} yields $Y_p \approx 0.244$, $D/H \approx 2.51 \times 10^{-5}$ ($z \approx -0.5$ vs. Coc [1]), ${}^3\text{He}/H \approx 1.01 \times 10^{-5}$, and ${}^7\text{Li}/H \approx 2.54 \times 10^{-10}$ (pre-astrophysical-depletion; $\approx 45\%$ below Coc semi-analytic 4.65×10^{-10} at the same η ; within the Spite-plateau envelope $(1.6\text{--}4.5) \times 10^{-10}$; `data/bbn_integrator.json`). Not claimed: derivation of η_{paper} from first principles in this note.

Patch QFT.. The quantum field theory used here is the HQIV *patch QFT*: fields, actions, charges, amplitudes, and readouts live on finite horizon patches labelled by shell indices $m \in \mathbb{N}$, finite spacetime index types such as `Fin 4`, and discrete carrier charts. Each patch is the *entire accessible universe* in theory terms—the full causally accessible patch net, not a mesh tile waiting for a fundamental continuum limit. This is the operative QFT of the model. Its empirical content is the finite-patch calculation itself: mass and coupling ladders, curvature ratios, shell thermodynamics, anomaly traces, branching ledgers, and rational witnesses are computed on the patch carrier and then compared with laboratory observables. Continuum notation, when used, is a dictionary for readers trained in smooth field theory; it is not the definition of the theory.

Gauge and topology on the patch carrier. The patch QFT has no hidden continuum gauge sector whose consistency must be assumed. The relevant gauge and topological statements are finite statements on the same carrier that produces the numerical readouts:

- **Anomaly cancellation.** For one Standard-Model generation $(Q_L, u^c, d^c, L, e^c, \nu^c)$, the patch trace identities give zero $U(1)_Y^3$, gravitational- $U(1)_Y$, $SU(3)_c^2 U(1)_Y$, $SU(2)_L^2 U(1)_Y$,

and $SU(3)_c^3$ anomaly coefficients; the $SU(2)_L^3$ coefficient vanishes because the weak doublet slot is pseudoreal. Three generations are a finite $\text{Fin } 3$ label sum of the same cancelled one-generation block, not a fitted multiplicity.

- **Topological sector.** The patch abelian gauge datum has a single topological sector. Instanton number, Pontryagin number, first Chern number, and $U(1)$ winding number vanish identically on patch data; the θ contribution is therefore θ -independent; and abelian commutator obstructions vanish.
- **Non-abelian charts and holonomy.** Light-cone $\mathfrak{so}(8)$ Lie closure, $G_2 \cup \{\Delta\}$ data, admissible holonomy on integrable Hopf shells, cyclic plaquette flatness, and weak/colour commutator certificates are discharged on the octonion carrier (see the sector zeta and holonomy discharge bundle in the Lean catalog). ****Matrix Wilson layer:**** full four-edge $SO(8)$ generator transport and Hopf-shell T^4 torsion plaquettes with pinned non-abelian witness ([Hqiv/Physics/SO8PlaquetteHolonomy.lean](#), witness `so8PlaquetteHolonomyDischarged_holds`). ****Wilson-kinetic equivalence**** ([Hqiv/Physics/ActionHolonomyGlue.lean](#), witness `wilsonKineticPlaquetteEquivalence_discharged`). ****Partial / open:**** Haar measure on rotated charts ([Hqiv/Physics/NonAbelianHolonomyMeasureScaffold.lean](#)); general flat HQVM kinetic boost and full strong-interacting discrete-action Poincaré ([Hqiv/Physics/DiscreteActionStrongPoincareBridge.lean](#), [Hqiv/Physics/DiscreteActionPoincareScaffold.lean](#)). Where the optional strong-colour certificate is invoked, the global Gell-Mann bracket law is proved on the finite $\mathfrak{su}(3)$ chart.
- **Rapidity phase and Lorentz invariance.** The null-lattice carrier `NullLatticeEvent` charts into 1+1 Minkowski space by the classical rapidity parametrization $\Lambda(\eta)(1, 1)$; rapidity addition on the carrier is equivariant under `boostMatrix11`, the Minkowski pairing is bilinearly invariant, and the polar-angle / `zetaHQIVTerm` phase is unchanged when ϕ transforms as a Lorentz scalar. The honest 3+1 extension preserves `minkowskiSq4` on the embedded $(t, x^1, 0, 0)$ plane.
- **Spatial rotations and directional readouts.** Orthogonal $O(3)$ rotations on $\text{Fin } 3$ spatial vectors preserve Euclidean inner products, $\|\mathbf{r} \times \mathbf{v}\|^2$, axis-projection fractions, and the spatial block of `minkowskiSq4` when time is fixed ([Hqiv/Geometry/SpatialRotationLorentzClosure.lean](#), witness `spatial_rotation_lorentz_closure_discharged`). Applied discharge: CMB sky rotation shifts the Stokes angle additively while preserving $\cos^2 + \sin^2 = 1$; O-Maxwell fluid sources rotate covariantly with gradients; coronal axis projections are joint rotation invariants. Full patch Lorentz on the flat chart is the rapidity bundle plus this spatial bundle (witness `full_lorentz_closure_discharged`). Continuum interacting QFT / Wightman theorems are not claimed.

External continuum theorems not required. Several textbook continuum constructions are *not* load-bearing prerequisites because the patch carrier already fixes the traces, holonomy transport, topological charges, and sector zeta bodies used for predictions: BPST instanton classification on a separate smooth four-manifold, Nielsen-style principal-bundle completeness on an external C^∞ base, and path-integral measure choice on that base. HQIV does not claim to re-prove those external theorems; it proves the finite patch statements that supply the same consistency checks for the readout pipeline. This is why familiar QFT language can appear in the comparison layer while the predictions remain patch-level statements with fewer adjustable inputs than a continuum effective fit.

Projection geometry and the classical ζ dictionary (2026-06). HQIV uses *zeta* in two distinct, now formally separated roles:

- **Sector zeta on the patch carrier** (TUFT+SM synthesis): fibre-torsion subleading bodies, `zetaHQIVTerm` phase on Hopf shells, and holonomy discharge bundles on the octonion light-cone carrier—finite spectral readouts indexed by shell and Fano sector, not the Riemann function.

- **Classical $\zeta(s)$ as an $\text{SO}(4)$ projection readout** ([14]): the Mathlib Riemann zeta packaged as a *regional* closed form on $\mathbb{C}$ —Dirichlet/Euler for $\Re s > 1$, sin-cos- Γ assembly on the strip, Bernoulli slots—with the functional-equation pair $(\sigma, 1 - \sigma)$ interpreted as a 45° quaternionic projection. Machine-checked on this spine: harmonic divergence forces Δ and $\mathfrak{so}(4)$ closure; shell sum $= \zeta(s)$ for $\Re s > 1$; polar decoupling $n^{-s} = n^{-\sigma} \cdot e^{-it \log n}$ (modulus σ -only, phase t -only); prime-log Diophantine independence; Goldbach circle $pq + (q - N)^2 = N^2$ as additive curvature in the log channel; and a large family of RH-*equivalent* locators (twiddle, spectral weights, polar collapse, transformer adjoint law, S^7 torsion cancellation) with *zero slack*—not a discharge of RH or Goldbach parity.

Continuum heat-flow and de Bruijn–Newman notation in the same story spine are *comparison ports* (`dbnHeatFamily`, lattice dominance hypotheses); they are not load-bearing continuum extensions of the patch. Formal mathematical notes may include only the readout-dictionary paragraph (`readout_dictionary_messaging.tex`) rather than this full contract when the manuscript is projection geometry rather than patch phenomenology.

HQIV in five sentences. (1) Cosmology is encoded on a discrete null-shell ladder $m(T) = \lfloor T_{\text{Pl}}/T \rfloor - 1$, not as a continuum mesh awaiting a limit. (2) Matter and gauge couplings read from an inner-outer Casimir chart $\kappa_6(\xi) = \eta_{\text{local}} \gamma C_2(\xi)$ tied to the Hopf–Beltrami mass bridge [6]. (3) Nucleon masses and cluster binding Q follow the eight-dimensional composite trace with a single proton anchor at referenceM = 4. (4) Baryon asymmetry η_{paper} is fixed upstream by curvature lock-in [5], with $\eta_{\text{BBN}} = \eta_{\text{paper}}$ proved in Lean. (5) BBN abundances are *weighted network readouts* on that ladder—this note certifies what is already theorem-level and documents the faithful Python integrator that closes Y_p , D/H , ${}^3\text{He}/H$, and ${}^7\text{Li}/H$ at η_{paper} .

Notation. We write η_{paper} for the lock-in baryon-to-photon ratio imported from baryogenesis; Lean proves $\eta_{\text{BBN}} = \eta_{\text{paper}}$ (no separate BBN fit). Deuteron, helium, and lithium binding enter as Q_D , $Q_{{}^4\text{He}}$, $Q_{{}^3\text{He}}$, $Q_{{}^7\text{Be}}$, $Q_{{}^7\text{Li}}$ from the composite-trace network unless a temperature-dependent release suffix $Q^{\text{eff}}(T)$ is explicit. Strong-channel weight $w_{\text{strong}} = 4/8$ and horizon ratio $\gamma = 2/5$ are HQIV constants, not BBN fit parameters.

1. SCOPE AND CLAIM DISCIPLINE

The HQIV programme ties discrete null-shell counting, horizon ratios, and octonionic-sector packaging [7, 11]. The baryogenesis note [5] fixes how $\eta(n; N)$ shares the same normalisation as $\Omega_k(n; N)$ at lock-in. **This note answers: given that normalisation, what light-element statements are already theorems in Lean, and what remains interpretive or comparison-layer physics?**

- (1) **Proved in Lean (network spine).** Valley counts for $A = 2, 3, 4$; cluster binding Q for deuterium and ${}^4\text{He}$ aligned with composite-trace nucleon binding; neutron fraction and $Y_p = 2x_n/(1 + x_n)$ in $(0, 1)$; η at BBN equals η_{paper} (the [BBN- \$\eta\$ identity](#)); positivity of D/H , ${}^3\text{He}/H$, ${}^7\text{Li}/H$ readouts at lock-in (the [network vital readout](#)).
- (2) **Proved in Lean (epoch spine).** Shell index $m(T) = \lfloor T_{\text{Pl,MeV}}/T \rfloor - 1$; BBN mid-epoch shell strictly above lock-in and below today (the [mid-epoch shell ordering](#)); epoch readout positivity at $T \approx 0.1 \text{ MeV}$ (the [epoch vital readout](#)).
- (3) **Proved in Lean (integrated witness).** Freeze-out Y_p in the Cooke-style observed band; integrated Y_p within 1% of freeze-out at exported numerics (the [integrated witness vital readout](#)).
- (4) **Comparison layer only.** Coc et al. [1] semi-analytic $Y_p(\eta, N_\nu, \tau_n)$ and $D/H(\eta)$ in the [BBN umbrella module](#)—not derived from HQIV weights.
- (5) **Not claimed.** Full 400-reaction PRIMAT network in Lean; Li-problem solution; independent refit of η or nucleon masses; continuum transport of weak rates.

2. UPSTREAM INPUTS CONSUMED AT LOCK-IN

The BBN stack does not introduce new dimensionful anchors. It consumes:

- **Baryon asymmetry.** η_{paper} from the baryogenesis witness chain [5], with $\eta_{\text{BBN}} = \eta_{\text{paper}}$ proved in the [network-from-weights module](#).
- **Nucleon masses.** `derivedProtonMass` and `derivedDeltaM` (neutron–proton gap) from the mass-spectrum lock-in discipline [6, 8]; binding shell `bbnBindingShell` = `referenceM`.
- **Composite-trace binding.** Deuterium and ^4He reaction Q values identified with cluster binding from the eight-dimensional composite trace (the [cluster-binding alignment theorems](#)).

Laboratory abundances and semi-analytic fits are *comparison targets* for scripts and narrative; they are not simultaneous fit rows in the proof graph.

3. NETWORK READOUT FROM VALLEY WEIGHTS

The [network-from-weights module](#) encodes light elements as a weighted readout on the same Fano/network picture used for nucleon binding [6]:

- **Valley counts.** $A = 2 \mapsto 2$, $A = 3 \mapsto 4$, $A = 4 \mapsto 6$ (helium-4 valley).
- **Boltzmann factor.** $\exp(-Q/T)$ at temperature T (MeV) for each channel Q .
- **Neutron sector.** $x_n = n/p$ from Δm and T ; $Y_p = 2x_n/(1 + x_n)$.
- **Light ions.** $D/H \propto \eta_{10}^{-\alpha_D} \times \text{thermal sink} \times \text{Boltzmann weight}$; analogous exponent slots for ^3He and ^7Li .

At lock-in temperature the readout packages $(\eta, Y_p, D/H, ^3\text{He}/H, ^7\text{Li}/H)$ with $\eta = \eta_{\text{paper}}$. The [network vital readout](#) theorem bundles positivity and the η identity—this is the certificate-level content of “BBN from network weights,” not a hand-tuned abundance table.

Remark 3.1 (Epoch dependence). *Partition formulas evaluated at lock-in temperature are not the same as relic abundances after integration over the BBN window; Section 4 and Section 5 separate partition readouts from cooling-integrated witnesses.*

4. BBN AS A COSMOLOGICAL EPOCH ON THE TEMPERATURE LADDER

The [epoch evolution module](#) maps temperature to shell index

$$m(T) + 1 = T_{\text{Pl,MeV}}/T,$$

with a QCD onset shell, lock-in shell $m_{\text{lockin}} = \text{referenceM}$, BBN window $T \in [0.01, 1] \text{ MeV}$, and CMB today as the cold end of the ladder. Proved ordering:

$$m_{\text{lockin}} < m_{\text{BBN,mid}} < m_{\text{now}},$$

where $m_{\text{BBN,mid}}$ corresponds to $T \approx 0.1 \text{ MeV}$. Freeze-out internal temperature is defined consistently with neutron–proton equilibrium break (the [freeze-out temperature identity](#)).

Remark 4.1 (Discrete shells and freeze-out sharpness). *Because $m(T)$ uses a floor map, the temperature ladder is piecewise constant in shell index: a cooling step crosses a shell boundary when T drops below $T_{\text{Pl}}/(m + 2)$. Freeze-out is therefore a shell predicate ($\tau_{n \leftarrow p} \sim t_{\text{exp}}$ at the matching shell), not a smooth continuum crossover. The integrator uses log-spaced sub-steps inside the BBN window to approximate the cooling history; residual $\mathcal{O}(\Delta m)$ artifacts near boundaries are absorbed into the documented D/H residual rather than hidden by a continuum transport code.*

The [epoch network module](#) supplies a rate scaffold (weak, np , dd , ^3He branches) and Hubble dilution—schematic reaction bookkeeping for integration witnesses, not a published PRIMAT clone in Lean.

4.1. Dynamic C_2 lapse clock (BBN lab).

Physical picture. In standard BBN one often parametrizes departures from freeze-out equilibrium via $H(T)$, τ_n , or effective reaction rates. HQIV instead advances species by a **shell reaction opportunity** $\mathcal{O} \propto \Delta \log \xi \times B_{\text{curv}}(\xi)$, where $\xi = T_{\text{Pl}}/T$ is the cosmological shell coordinate [5]. The Hopf–Beltrami mass bridge packages the matter slot as $\kappa_6(\xi) = \eta_{\text{local}} B_{\text{curv}}(\xi) \gamma C_2(\xi)$ (Hopf–Beltrami mass bridge). On the hot BBN chart B_{curv} is nearly flat while lapse concentration $C_2(\xi)$ grows with ξ ; the anchor-normalized ratio $\kappa_{6,\text{ref}}/\kappa_6(T)$ therefore acts as a *local lapse reaction clock*: it is **not** a variable speed of light in the metric sense, but an effective rescaling of strong-channel opportunity relative to the freeze-out anchor temperature $T_{\text{freeze}}(\eta)$. In Friedmann language, HQIV does not import $H(T)$ as the integration clock; the dictionary is “comoving temperature step $\Delta \log T$ ” $\leftrightarrow$ “shell opportunity $\Delta \log \xi$ ” modulated by $(\kappa_{6,\text{ref}}/\kappa_6)^{w_{C_2}(T)}$ in the deuterium bottleneck. This is closest to a **temperature-dependent effective rate** in the bottleneck window $T \leq T_{\text{bn}}(\eta)$, with exponent tied to composite-trace binding release rather than a fitted MeV cutoff.

Definitions. For $T \leq T_{\text{bn}}(\eta) = \gamma w_{\text{strong}} T_{\text{freeze}}(\eta)$,

$$\begin{aligned} w_{C_2}(T) &= \gamma w_{\text{strong}} \frac{Q_D^{\text{eff}}(T)}{Q_{np}}, \\ f_{C_2}(T) &= \left(\frac{\kappa_{6,\text{ref}}}{\kappa_6(T)} \right)^{w_{C_2}(T)}, \\ \kappa_6(T) &= \eta_{\text{paper}} B_{\text{curv}}(\xi(T)) \gamma C_2(\xi(T)). \end{aligned}$$

Lean names: `bbnDynamicC2BottleneckT_MeV`, `bbnDynamicC2LapseExponent`, `bbnDynamicC2OpportunitySuppression` (**DynamicBBNBaryogenesis**). Multiplying opportunity by raw κ_6 at low T *over-suppresses* D/H ; the inverse lapse ratio is the certified dynamic- C_2 readout.

The legacy dynamic-bulk integrator (`hqiv_dynamic_bulk_bbn.py`) implements this opportunity clock without the synthesis-window stoichiometry of Section 4.2.

Provider decomposition at η_{paper} . Table 1 quantifies how each layer raises D/H toward the faithful readout (exported by `hqiv_bbn_paper_tables.py`). Stoichiometric $n+p \rightarrow D$ capture and the synthesis window ($1 \text{ MeV} \rightarrow T_{\text{bn}}$) dominate ($4.4 \times 10^{-6} \rightarrow 1.84 \times 10^{-5}$); coupled inventory with outside-neutron curvature supplies the final increment to 2.51×10^{-5} and sets Y_p via the post-synthesis x_n survivor. Turning off C_2 lapse alone leaves the faithful D/H unchanged at this precision— f_{C_2} is booked primarily on the $n+p$ and $D+D$ branches inside the bottleneck, not as a separate post-hoc multiplier.

TABLE 1. Layered D/H providers at $\eta_{\text{paper}} = 6.10 \times 10^{-10}$ (`data/bbn_paper_tables.json`).

Provider layer	D/H	Y_p
Lock-in Q_D at T_{freeze}	$\lesssim 10^{-30}$	—
Hybrid $Q^{\text{eff}}(T)$, partition window	4.4×10^{-6}	0.258
+ stoichiometric + C_2 , full window	8.1×10^{-6}	0.247
+ synthesis window (uncoupled)	1.84×10^{-5}	0.247
Faithful (coupled inventory + curvature)	2.51×10^{-5}	0.244
Coc et al. comparison [1]	$(2.53 \pm 0.04) \times 10^{-5}$	0.244

4.2. Faithful stoichiometric integrator. The recommended reproduction path is Section C (`hqiv_bbn_integrator.py`), mirrored in Lean by the **stoichiometric integrator module** and `data/bbn_integrator.json`. At $\eta_{\text{paper}} = 6.10 \times 10^{-10}$ it couples:

- **Synthesis-window D/H .** Log-weighted integration from 1 MeV through the C_2 bottleneck ($T_{\text{bn}} \approx 0.14 \text{ MeV}$), not the full cold tail to 0.01 MeV. Below the ${}^3\text{He}$ synthesis mid-gate

($T_{\text{peak}} = 0.18 \text{ MeV}$, Lean bbnHe3SynthesisMidMeV) the log weights taper linearly $T_{\text{bn}} \rightarrow T_{\text{peak}}$ via bbnSynthesisDWindowTailGate—a thermal-crush tail, not a hard shut-off at mid-MeV. The numerator adds an explicit $n+p \rightarrow D$ capture term (`branch_np_to_d`) and a strong-channel deposition boost $\propto \gamma w_{\text{strong}}(1 + g_{np} + c_2)$ (bbnSynthesisStrongDepositionBoost).

- **Coupled neutron inventory.** Outside-curvature and ν -opacity set $\tau_n(T)$; the same $x_n(T)$ ladder scales $n+p \rightarrow D$ and $D+D \rightarrow {}^4\text{He}$. Y_p is read from the *post-synthesis* survivor $x_n(T_{\text{low}})$, not the instant freeze-out partition.
- **${}^3\text{He}/H$ window.** Dedicated $0.45 \rightarrow 0.05 \text{ MeV}$ ladder with trimer-width erosion on the strong channel only; a narrower feed gate routes ${}^3\text{He}$ into the ${}^7\text{Be}$ tail without inflating the ${}^3\text{He}$ readout.
- **${}^7\text{Li}/H$ tail.** ${}^7\text{Be}$ production and electron capture on $0.25 \rightarrow 0.05 \text{ MeV}$; witness ${}^7\text{Li}/H \approx 2.54 \times 10^{-10}$ (pre-depletion HQIV readout).

Exported readout: $Y_p \approx 0.244$, $D/H \approx 2.51 \times 10^{-5}$ ($z \approx -0.5$ vs. $(2.53 \pm 0.04) \times 10^{-5}$), ${}^3\text{He}/H \approx 1.01 \times 10^{-5}$, ${}^7\text{Li}/H \approx 2.54 \times 10^{-10}$ ($\approx 45\%$ below Coc 4.65×10^{-10}). Curvature relative to bare equilibrium *lowers* Y_p by $\sim 3 \times 10^{-3}$ while *raising* D/H slightly—the intended unified inventory behaviour, not a Y_p -only tweak.

5. INTEGRATED WITNESSES AND PYTHON EXPORT

The **BBN witness module** pins numeric exports from `data/bbn_witnesses.json` (generated by `scripts/hqiv_bbn_abundances.py` and `scripts/hqiv_bbn_epoch_network.py`; Appendix D). At the exported lock-in $\eta = 6.10 \times 10^{-10}$ (partition readout in `data/bbn_witnesses.json`):

- Lock-in cluster Q values from the composite trace: $Q_D \approx 2.34 \text{ MeV}$, $Q_{{}^4\text{He}} \approx 7.03 \text{ MeV}$, $Q_{{}^3\text{He}} \approx 4.39 \text{ MeV}$; post- α wells $Q_{{}^7\text{Be}} \approx 14.9 \text{ MeV}$, $Q_{{}^7\text{Li}} \approx 14.4 \text{ MeV}$ (${}^7\text{Be}$ deeper than ${}^7\text{Li}$).
- Integrated window $Y_p \approx 0.247$ lies in the Cooke et al. band checked in Lean ([freeze-out \$Y_p\$ band witness](#)).
- Integrated Y_p is within 1% of the freeze-out partition value ([integrated-freeze-out proximity witness](#)).
- Faithful stoichiometric integrator at η_{paper} : $Y_p \approx 0.244$, $D/H \approx 2.51 \times 10^{-5}$, ${}^3\text{He}/H \approx 1.0 \times 10^{-5}$, ${}^7\text{Li}/H \approx 2.54 \times 10^{-10}$ (`data/bbn_integrator.json`; §4.2).
- Legacy dynamic-bulk row (`legacy_dynamic_bulk_row` in `integrator_lean_audit.json`): pre-synthesis-window scaffold only; superseded by `faithful_bbn_integrator`.

Honesty on lithium and deuterium. The faithful integrator exports ${}^7\text{Li}/H \approx 2.54 \times 10^{-10}$ (pre-astrophysical-depletion) at lock-in η , $\approx 45\%$ below the Coc semi-analytic comparison 4.65×10^{-10} but inside the broad Spite envelope $(1.6\text{--}4.5) \times 10^{-10}$, without refitting nuclear Q values. D/H at η_{paper} sits within $\sim 0.5\sigma$ of the Coc anchor once the synthesis-window tail gate (bbnSynthesisDWindowTailGate) is booked—documented as a structural readout, not a comparison-layer fit. Mid-epoch partition witnesses in `bbn_witnesses.json` can still diverge on Li; treat those as epoch-scaffold exports distinct from the stoichiometric integrator.

6. RELATION TO STANDARD BBN PHYSICS

Neutron lifetime and weak rates. Laboratory $\tau_n \simeq 879.4 \text{ s}$ enters as a **reference scale** in [NuclearOutsideTemperatureDynamics](#) and the Python outside-neutron module; it is **not refit** at BBN. Instead HQIV books a temperature-dependent ratio $\tau_n(T)/\tau_{n,\text{lock}}$ from (i) outside-curvature support on the free neutron branch and (ii) relic-neutrino opacity catalysis on the weak width (same stack as nucleon β tipping [4]). At T_{freeze} the witness exports $\tau_n(T)/\tau_{n,\text{lock}} \approx 1.19$ and a weak-width factor ≈ 1.07 (`data/bbn_paper_tables.json`). This lengthens the effective freeze-out epoch slightly and feeds the coupled $x_n(T)$ march that sets both Y_p and D/H .

Effective neutrino species N_ν . HQIV does not introduce an explicit N_ν fit parameter in the proof graph. The epoch scaffold assumes the Standard Model relativistic neutrino content implicitly through the thermal g_* bookkeeping used in comparison-layer Coc fits [1]. A future extension would route N_ν through the same relic- ν opacity slot already present in the outside-neutron module.

Baryon density η_{paper} vs. Planck/CMB.. $\eta_{\text{paper}} = 6.10 \times 10^{-10}$ ($\eta_{10} = 6.10$) is the Lean lock-in anchor imported from baryogenesis [5], matching the BBN+CMB comparison target 6.10 ± 0.06 recorded there—not a simultaneous abundance fit. The dynamic bulk integrator independently derives $\eta_{10} \approx 6.20$ from Ω_b and curvature closure; the faithful BBN readout is evaluated at the proved identity $\eta_{\text{BBN}} = \eta_{\text{paper}}$.

The lithium problem. Standard SBBN with unmodified rates typically *overpredicts* ${}^7\text{Li}/H$ by factors $\sim 2\text{--}3$ relative to the Spite plateau once post-BBN destruction is accounted for. At η_{paper} Coc semi-analytics [1] give ${}^7\text{Li}/H \approx 4.65 \times 10^{-10}$; the HQIV faithful witness is ${}^7\text{Li}/H \approx 2.54 \times 10^{-10}$ (pre-astrophysical-depletion), $\approx 45\%$ below that comparison anchor but inside the broad $(1.6\text{--}4.5) \times 10^{-10}$ envelope, without refitting Q . Table 2 shows the mechanism: unstratified ${}^3\text{He}$ routing inflates ${}^7\text{Be}$ production by orders of magnitude; the combination of **trimer-width erosion** on the ${}^3\text{He}$ readout and a **narrow** ${}^3\text{He} \rightarrow {}^7\text{Be}$ **feed gate** (0.05–0.35 MeV) brings the ladder into band. Composite-trace $Q_{{}^7\text{Be}}$ sets the Boltzmann weight; C_2 lapse affects D/H more strongly than ${}^7\text{Li}$ in the final witness.

TABLE 2. ${}^7\text{Li}/H$ sensitivity to ${}^3\text{He}$ stratification gates at η_{paper} (`bbn_paper_tables.json`).

${}^3\text{He}$ stratified	synthesis window	${}^7\text{Li}/H$
off	off	3.1×10^{-7}
off	on	4.4×10^{-10}
on	off	1.8×10^{-9}
on	on	2.54×10^{-10}

7. QUANTITATIVE VALIDATION AND UNCERTAINTIES

Y_p band used in witnesses. Lean **freeze-out Y_p band witness** checks against center 0.244 ± 0.004 (Cooke-style helium band; [1]), **not** the wider 0.244–0.256 envelope sometimes quoted from heterogeneous datasets. The faithful integrator readout $Y_p \approx 0.2442$ has $z \approx +0.05$ against that witness; the partition freeze-out witness 0.2469 remains inside the same band.

D/H and observational context. The model $D/H \approx 2.51 \times 10^{-5}$ lies within $\sim 0.5\sigma$ of the Coc semi-analytic anchor $(2.53 \pm 0.04) \times 10^{-5}$ and the high-precision deuterium measurements of Cooke et al. [2, 3]. The log-weight tail below $T_{\text{peak}} = 0.18$ MeV (shared with the ${}^3\text{He}$ synthesis mid-gate) closes the prior $\sim 6\%$ undershoot that arose from treating the synthesis window as a hard floor at 0.15 MeV. Recent “high- D ” interstellar sightlines and galactic chemical evolution (GCE) depletion corrections remain an active comparison layer: HQIV documents offsets at η_{paper} rather than tuning γ , Q_D , or η to match Coc. At $\eta_{10} = 5.8$ the same integrator reaches $D/H \approx 2.79 \times 10^{-5}$, bracketing the Coc value from above.

Sensitivity to HQIV constants. Table 3 summarizes one-at-a-time variations around the proved constants $\gamma = 2/5$ and $w_{\text{strong}} = 4/8$. D/H scales roughly linearly with γ and w_{strong} in the synthesis window; Y_p is comparatively stable because it is anchored by the post-synthesis x_n march.

η sweep vs. comparison layer. Table 4 lists the faithful integrator across $\eta_{10} \in [5.8, 6.4]$. Coc semi-analytics [1] are not inputs; the table is for external calibration only.

8. COMPARISON LAYER: COC ET AL. SEMI-ANALYTICS

The **BBN umbrella module** records Coc et al. (2015) semi-analytic $Y_p(\eta, N_\nu, \tau_n)$ and $D/H(\eta)$ for side-by-side plots. These definitions are explicitly **not HQIV inputs**; they exist so reviewers can see

TABLE 3. Sensitivity of faithful readout to γ and w_{strong} (one-at-a-time; `bbn_paper_tables.json`).

Parameter	value	D/H	Y_p
γ	0.36	1.95×10^{-5}	0.2447
	0.40	2.51×10^{-5}	0.2442
	0.44	3.23×10^{-5}	0.2437
w_{strong}	0.45	1.95×10^{-5}	0.2447
	0.50	2.51×10^{-5}	0.2442
	0.55	3.23×10^{-5}	0.2437

TABLE 4. Faithful integrator abundances vs. η_{10} (`bbn_paper_tables.json`).

η_{10}	Y_p	D/H	${}^3\text{He}/H$	${}^7\text{Li}/H$
5.8	0.254	2.79×10^{-5}	1.08×10^{-5}	2.83×10^{-10}
5.9	0.251	2.69×10^{-5}	1.05×10^{-5}	2.73×10^{-10}
6.0	0.247	2.60×10^{-5}	1.03×10^{-5}	2.63×10^{-10}
6.1	0.244	2.51×10^{-5}	1.01×10^{-5}	2.54×10^{-10}
6.2	0.241	2.43×10^{-5}	9.9×10^{-6}	2.45×10^{-10}
6.3	0.238	2.35×10^{-5}	9.7×10^{-6}	2.36×10^{-10}
6.4	0.235	2.27×10^{-5}	9.5×10^{-6}	2.28×10^{-10}

how close the weight readout sits to standard BBN phenomenology without contaminating the proof graph. Cooke et al. deuterium and helium measurements [2, 3] supply independent observational anchors for the same comparison layer.

9. UMBRELLA CERTIFICATE

The `BBN umbrella module` conjoins network, epoch, and integrated vital readouts into `bbn_full_vital_readout`. This is the single top-level Prop a downstream cosmology or CMB paper can cite once lock-in η and nucleon witnesses are accepted from earlier tiers.

10. WHAT THIS NOTE DOES NOT CLAIM

- Derivation of η_{paper} from associator phases or sphaleron yields (baryogenesis interpretive layer).
- Replacement of PRIMAT/MESA with the HQIV rate scaffold.
- Full Lean certificate that integrated D/H matches Coc et al. within stated uncertainty at every η (Python witness is within $\sim 0.5\sigma$ at η_{paper} ; see §4.2).
- Post-BBN Li destruction and galactic chemical evolution (pre-depletion ${}^7\text{Li}/H$ only).
- Any continuum-limit or patch-refinement programme.

11. CONCLUSION

Big-bang nucleosynthesis in HQIV is formalised as a *weighted network readout* at lock-in, lifted to a *cosmological epoch* on the temperature-shell ladder, and closed by *integrated abundance witnesses* tied to Python exports. The proof graph is sorry-free at the vital-bundle level; observational agreement is claimed only where the witness module states it (Y_p in the fixed- η bundle; light-ion band agreement in the faithful stoichiometric witness of §4.2, including D/H within $\sim 0.5\sigma$ of Coc at η_{paper}). Companion reproduction is documented in Appendix D.

ACKNOWLEDGMENTS

Machine-checked proofs live in the public HQIV Lean repository [15]. BBN witness JSON is regenerated from the scripts named in Appendix D.

APPENDIX A. LEAN MODULE INDEX

Each entry lists the repository path (audit) and the reader-facing label used in the body. All modules are in the `Hqiv.Physics` namespace unless noted.

- : **BBN umbrella** [Hqiv/Physics/BigBangNucleosynthesis.lean](#) — Coc comparison defs; `bbn_full_vital_readout`.
- : **Network from weights** [Hqiv/Physics/BBNNetworkFromWeights.lean](#) — Valley counts, Y_p , D/H , η identity; `bbn_network_vital_readout_holds`.
- : **Epoch evolution** [Hqiv/Physics/BBNEpochEvolution.lean](#) — Shell ladder, epoch readout; `bbn_epoch_vital_readout_holds`.
- : **Epoch network** [Hqiv/Physics/BBNEpochNetwork.lean](#) — Rate scaffold, cooling integration state.
- : **BBN witness** [Hqiv/Physics/BBNWitness.lean](#) — JSON-linked Y_p witnesses; `bbn_integrated_witness_vital_holds`.
- : **Network vital readout** `bbn_network_vital_readout` in [Hqiv/Physics/BBNNetworkFromWeights.lean](#).
- : **Epoch vital readout** `bbn_epoch_vital_readout` in [Hqiv/Physics/BBNEpochEvolution.lean](#).
- : **Integrated witness vital** `bbn_integrated_witness_vital` in [Hqiv/Physics/BBNWitness.lean](#).
- : **Full BBN vital readout** `bbn_full_vital_readout` in [Hqiv/Physics/BigBangNucleosynthesis.lean](#).
- : **BBN- η identity** `eta_bbn_eq_eta_paper` in [Hqiv/Physics/BBNNetworkFromWeights.lean](#).
- : **Mid-epoch shell ordering** `bbnMidEpochShell.gt.lockinShell`, `bbnMidEpochShell.lt.nowShell` in [Hqiv/Physics/BBNEpochEvolution.lean](#).
- : **Freeze-out temperature identity** `bbnInternalTemperatureMeV.eq.freezeout` in [Hqiv/Physics/BBNEpochEvolution.lean](#).
- : **Cluster-binding alignment** `bbnDeuteronBindingQ.eq.clusterBinding`, `bbnHelium4BindingQ.eq.clusterBinding` in [Hqiv/Physics/BBNNetworkFromWeights.lean](#).
- : **Freeze-out Y_p band witness** `bbnYpFreezeoutWitness.in_observed_band` in [Hqiv/Physics/BBNWitness.lean](#).
- : **Integrated-freeze-out proximity** `bbnIntegratedYp_near_freezeout_witness` in [Hqiv/Physics/BBNWitness.lean](#).
- : **Nuclear valley scaffold** [Hqiv/Physics/HQIVNuclei.lean](#) — Isotope valley counts feeding BBN weights.
- : **Lock-in η paper module** [Hqiv/Physics/BaryogenesisEtaPaper.lean](#) (upstream).
- : **Derived nucleon mass** [Hqiv/Physics/DerivedNucleonMass.lean](#) (upstream).
- : **BBN dynamic- C_2 pack** [Hqiv/Physics/DynamicBBNBaryogenesis.lean](#) — `bbnDynamicC2OpportunitySuppression`, `bbnShellReactionOpportunity_dynamic_integrator`; [Hqiv/Physics/HopfShellBeltramiMassBridge.lean](#) — `tuftLapseConcentrationAtXi`, `tuftHopfKappa6AtXi`.
- : **BBN stoichiometric integrator** [Hqiv/Physics/BBNStoichiometricIntegrator.lean](#) — synthesis-window D/H with `bbnSynthesisDWindowTailGate`, `bbnHe3SynthesisGate`, `bbnSynthesisStrongDepositionBoost`, coupled neutron inventory; Python mirror `hqiv.bbn.integrator.py`.
- : **Outside neutron dynamics** [Hqiv/Physics/NuclearOutsideTemperatureDynamics.lean](#) — free-neutron $\tau_n(T)$ modulator feeding the faithful integrator.

APPENDIX B. STOICHIOMETRIC INTEGRATOR FLOW

Reader-facing pseudocode for `hqiv.bbn_integrator.py` (Lean mirror: [stoichiometric integrator module](#)):

```

INPUT: eta_paper, lock-in Q from composite trace, gamma, w_strong
BUILD temperature ladder: T = 1 MeV -> T_bn (C2 bottleneck)
MARCH coupled neutron inventory x_n(T):
    tau_n(T) from outside curvature + relic-nu weak-width factor
    beta-decay survival freeze -> T_bn
FOR each T on ladder (log weights w):
    tail_gate(T) = bbnSynthesisDWindowTailGate(T) # unity >= T_peak=0.18 MeV
    w_eff = w * tail_gate(T)
    Q_D^eff(T), Q_3^eff(T) from binding release + trimer width
    f_C2(T) = (kappa6_ref/kappa6(T))^w_C2(T) if T <= T_bn else 1

```

```

D/H from stoichiometric branches:
  n+p->D capture (branch_np_to_d * f_C2)
  D+D->4He, D photodissociation, D+p->3He (width-suppressed)
  strong deposition boost ~ gamma * w_strong * (1 + g_np + c2)
  integrate D/H with sum(D/H(T) * w_eff) / sum(w_eff)
  3He/H on 0.45 -> 0.05 MeV window (separate ladder)
  7Be -> 7Li tail on 0.25 -> 0.05 MeV
OUTPUT Y_p = 2*x_n(T_low)/(1+x_n(T_low)) # post-synthesis survivor
        D/H, 3He/H, 7Li/H integrated abundances

```

APPENDIX C. MINIMAL REPRODUCTION AND SAMPLE WITNESS

From the repository root, a single command regenerates the primary witness:

```
PYTHONPATH=scripts python3 scripts/hqiv_bbn_integrator.py
```

Paper tables (Table 1, Table 4, Table 3):

```
PYTHONPATH=scripts python3 scripts/hqiv_bbn_paper_tables.py
```

Excerpt from data/bbn_integrator.json at η_{paper} :

```

"abundance_readout": {
  "Y_p": 0.24422,
  "D_over_H": 2.5119e-05,
  "He3_over_H": 1.0122e-05,
  "Li7_over_H": 2.5362e-10,
  "synthesis_d_T_peak_MeV": 0.18,
  "x_n_at_T_low": 0.13909,
  "delta_Y_p_curvature": -0.00269
}

```

APPENDIX D. REPRODUCER SCRIPTS

From the repository root:

```

PYTHONPATH=scripts python3 scripts/hqiv_bbn_integrator.py
PYTHONPATH=scripts python3 scripts/hqiv_bbn_paper_tables.py
python3 scripts/hqiv_bbn_abundances.py
python3 scripts/hqiv_bbn_epoch_network.py
python3 scripts/hqiv_dynamic_bulk_bbn.py
python3 scripts/hqiv_integrator_lean_audit.py
python3 -m unittest scripts.test_hqiv_bbn_integrator
python3 -m unittest scripts.test_hqiv_dynamic_c2_bbn

```

Outputs.

- data/bbn_integrator.json — primary abundance witness
- data/bbn_paper_tables.json
- data/bbn_witnesses.json
- data/dynamic_bulk_bbn_v2.json
- data/integrator_lean_audit.json

See papers/bbn/scripts/README.md for the patch-ontology disclaimer and input list.

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INDEPENDENT RESEARCHER

Email address: **steven@disregardfiat.tech**